

Migration, Networks, and Religious Choice*

Timothy Hersey[†]

November 4, 2024

[Click here for the latest version.](#)

Abstract

Over the last half century, affiliation to Pentecostal denominations has grown rapidly, with this growth concentrated in low income countries. This paper investigates one economic factor influencing this growth - rural-urban migration: religious institutions attract members by serving as hosts for economic and social networks, a service which is especially attractive for migrants into the city. I present new evidence from Ghana to test this hypothesis, using data from the ISSER Northwestern-Yale Panel Survey (GSPS) and a survey of Pentecostal churches in Accra. I construct a series of shift-share instruments for migration, using exposure to price shocks in the mining, construction and manufacturing industries. I find that migration significantly increases Pentecostal affiliation, especially for migrants with few network connections in the city. Pentecostal churches in urban areas specialise in the provision of network services; extracting more contributions and providing more transfers than their competitors. The process of migration also has implications for village networks. I find that out-migration induces conversion for households at the origin away from Catholicism. I present evidence consistent with the story that out-migrant households convert in response to weakening ties with the village network.

*I am indebted to my advisors, Mushfiq Mobarak, Mark Rosenzweig, Jose-Antonio Espin-Sanchez and Ben Polak, for their guidance throughout the many stages of this project. I also thank Lauren Falcao Bergquist, Kevin Donovan, Timothy Guinnane, Costas Meghir, Kaivan Munshi, Rohini Pande, Nicholas Ryan, Nate Barker, Alvaro Cox, Mirco Dinelli, Alicia Entem, Matthew Schwartzman, Jillian Stallman, and numerous seminar participants at Yale for their insightful comments.

[†]Yale University. Email: timothy.hersey@yale.edu

1 Introduction

Across the last century, there have been monumental shifts in the patterns of religious affiliation around the globe. In contrast to a steadily secularising West, religiosity in much of the developing world is climbing. One trend embedded in this transition is the rapid rise of new Christian Pentecostal denominations, especially within Africa and Latin America, rising from a near zero share in 1900 to 8% of the global population today (World Christian Database, 2024). These Pentecostal denominations dispense with many of the rituals and formality found within the mainline Catholic and Protestant denominations. Critical to understanding the forces that underlie this trend is diagnosing the good (in the economic sense) that Pentecostal churches are providing, and why this good might be especially valuable in a low income, rapidly urbanising context.

One key service provided by Pentecostal churches, and religious institutions more broadly, is functioning as hosts for social and economic networks (Chen, 2010; Ager and Ciccone, 2018; Murphy et al., 2019). Networks are critical in the process of migration, assisting in the integration of a migrant into a new economic environment (Beaman, 2012; Munshi, 2003); providing risk sharing services, and pathways to employment. As the process of urbanisation draws people into the city, the services provided by religious institutions will be especially valuable for migrants. Migration and urbanisation may, therefore, play a central role in shaping patterns of religious affiliation over time. If Pentecostal churches are particularly successful at facilitating economic networks, especially for migrants, increases in migration may induce waves of Pentecostal conversion.

The hypothesis explored in this paper is the following: Rural-Urban migration is an important factor in the growth of Pentecostalism, owing to the comparative advantage Pentecostal churches hold in facilitating economic and social networks. As migrants move from the village to the city, they look for a network to help them integrate into the urban environment. The network they find with the highest returns are those provided by Pentecostal churches, inducing conversion for these migrants. Furthermore, there may be a conversion effect for the household at origin. As the migrant converts, they may introduce their family at origin to a new religious good, leading to conversion. Alternatively, the process of outmigration

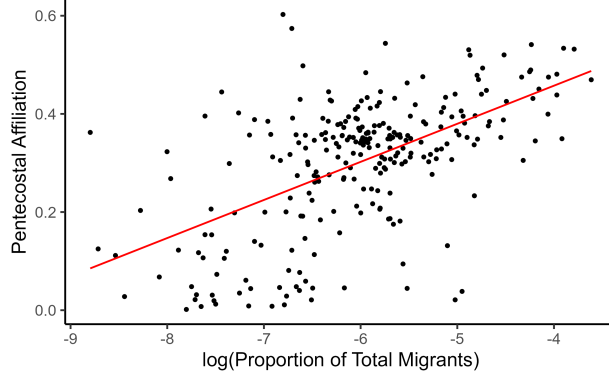


Figure 1: 2021 Ghana Census. Each point represents a district.

may break down village risk-sharing networks as in Munshi and Rosenzweig (2016); Morten (2019). If the households in that village value the services provided by the village network, they may search for these services from an alternate source, joining new religious institutions as network providers.

I test this hypothesis in the context of Ghana, whose Pentecostal transition reflects the experience of many African nations. Indeed, there is descriptive evidence of the relationship between migration and Pentecostal growth. Figure 1 shows the correlation between Pentecostalism and the proportion of total in migrants across districts. Districts receiving a higher share of migrants, have a higher share of Pentecostal affiliation. Table 1 reports regression results from the 2021 Ghana census on the correlation between migration and Pentecostal affiliation. Holding age, gender, and education fixed, migrants are 5.8% more likely to be Pentecostal, while rural-urban migrants are 9.2% more likely to be Pentecostal.

	<i>Dependent variable:</i>		
	<i>Pentecostal</i>		
	(1)	(2)	(3)
Migrate	0.058*** (0.003)		
Rur-Urb Migrate		0.092*** (0.004)	
Rur-Urb Migrate (5 Years)			0.062*** (0.003)
Mean	0.310	0.310	0.310
Observations	1,797,161	1,797,161	1,789,045
Adjusted R ²	0.114	0.117	0.114

Table 1: Includes fixed effect for district at birth. HH controls for age, gender, education. Errors clustered at the district level.
*p<0.1; **p<0.05; ***p<0.01,

There is also evidence that conversion is happening in rural areas - the highest growth in Pentecostalism over the last 20 years has occurred in rural districts in the east (see figure A.1).

The hypothesis predicts conversion both for migrants into urban areas, and for the families of migrants remaining in rural areas. The paper takes these two components in turn.

I begin by causally identifying the effect of migration on the religious affiliation of the migrant. To do this I construct a set of shift-share instruments, using shocks to prices for industries in potential destination districts as plausible exogenous variation. I focus on the manufacturing and construction industries as core low skill employers in urban areas, and the mining industry as an employer in rural areas, using shocks to these industries as a potential pull-shock for migration. Price shocks are multiplied by a form of market access to the industry - the sum of distances from origin districts to potential destinations weighted by the industry share in destination districts. This is multiplied by a second share - an individual productivity share, constructed using years of education and an age probability of migration. Using the full set of instruments I find that migration significantly increases Pentecostal affiliation for male migrants from rural areas.

I then compare potential explanations for the relationship between migration and religious affiliation. I present evidence that Pentecostal institutions provide network services in the form of monetary and in-kind transfers, both formally through the institution, and informally through the network of members. Pentecostal institutions also provide support for businesses, allowing members to advertise their businesses during services. I test for differences in this provision of network services across religious denominations, and find that the provision of services is significantly greater in Pentecostal churches. Given this function as network service providers, I look for evidence that migrants convert in pursuit of these services. I test if migrants with fewer network connections, i.e. migrants with greater returns from network engagement, are more likely to convert. Migrants who report not knowing anyone at their destination are significantly more likely to convert. Even with this evidence, it may be possible that either 1) migrants, especially unconnected migrants, are attracted to Pentecostal churches for the hopeful messages they present, or 2) while migrants do join Pentecostal churches for the network, they purely value the social good of the network. I

seek to rule out these stories by 1) comparing the beliefs and psychology of migrants who join, and 2) comparing employed and unemployed migrants, holding fixed their network connections.

To estimate the effect of out-migration on origin households, I repeat the instrument exercise described above. I find that outmigration induces changes in religious affiliation in the origin household. Surprisingly, in aggregate shifts mostly occur away from Catholicism towards Protestantism. Examining the services provided by religious institutions in rural areas, I find that Pentecostal and Protestant churches provide a similar scale of services in rural areas, and that religious service provision is generally rarer in rural areas than it is in urban. I test the network hypothesis for out-migrant households - that households convert as the village network breaks down, against alternative hypotheses - households convert based on the experience of the migrant. I estimate the effect of out-migration on connections with the village network, and look at the effect of the migrant's religious experience on the conversion of the remaining household. After out-migration events households engage less with village networks, and display less trust in the village network.

In presenting this evidence, I introduce two novel datasets. I firstly use the ISSER-Northwestern-Yale Long Term Ghana Socioeconomic Panel Survey (GSPS), which traces religious affiliation and migration status across the waves of the survey. I contribute a new religion unit to the fourth wave of the survey, allowing a mapping between the pre-existing detailed economic information of the survey, and information on religious participation, institutional membership, services received from religious institutions, and the history of religious affiliation.

I secondly present the results of a detailed survey of 35 Pentecostal institutions I conducted in Accra, containing information on institutional history, institutional organisation, membership, employees, finance, and service provision. While there is a rich literature relating migration to networks, this research examines a novel type of network in the context. Compared to other common networks, religious networks are not explicitly discriminatory - any individual has the opportunity to join, and are often encouraged to do so. In fact, religious institutions can provide unique opportunities to socialise across demographic groups (Chetty et al., 2022).

Having a formal organisation act as host has multiple benefits. Religious institutions act

as a coordinating device, providing the space and opportunity for a group to socialise and build trust with one another through repeat interactions. In addition to this coordination role, religious institutions can act to manage network services like consumption smoothing, pooling resources and allocating transfers to members with the greatest need, and referral networks, connecting members in need of a job or housing with potential employers or hosts. These roles may be especially important in urban areas, where the environment is rapidly changing, and there are interactions between many disparate groups.

For Pentecostal churches to benefit from migration patterns, they must uniquely be structured to provide services valuable to migrants. An effective network host is one that provides centralised, credible and timely services to their members, and extracts contributions fairly and effectively. Institutions that by nature are participatory, flexible, trustworthy and hold lower levels of bureaucracy will be most effective at fulfilling this role.

Pentecostal churches fill this brief in several ways. The traditional Protestant and Catholic denominations are large, historical organisations. The doctrine and practices of the organisations are set, and restrictions are placed on the types of services and activities that can occur through the institution. Pastors or Priests within these denominations must be trained for several years before practising. The use of finance must generally be approved by a parish board, and is subject to oversight from regional authorities. Wages are generally set by the denomination, and only loosely tied to the income of the local branch. In comparison, Pentecostal churches are largely informal and independent. In Ghana there are no formal registration requirements for churches and very few barriers to entry. Most Pentecostal churches are started locally, independent of a larger denominational body. Much of the organisational structure, therefore, is constructed by the founder. As a result, the decision making and processing of resources can become concentrated around this central figure. There are no formal training requirements for Pentecostal pastors, and Pentecostal churches are more reliant on volunteer hours from their members.

There are two main implications of the independence and informality of Pentecostal churches. Firstly, Pentecostal churches have the ability to innovate, both in the doctrine they present, and in the services they offer to their members, catering these goods more closely to local demand. In effect, the introduction of Pentecostal churches shifted the market from an

oligopoly market with a few large players, to a perfectly competitive market, with independent Pentecostal churches absorbing a large market share. Secondly, Pentecostal pastors are endowed with an incentive structure that promotes growth. Since the organisation is concentrated around the founder, and their profitability depends on the contributions of their members, the founder has greater incentives to attract new members to their congregation, and to extract greater commitments from existing members. Under a club goods model, this higher level of commitment allows more network services, and can increase the utility of members in the organisation. This incentive structure is stronger in more densely populated, and high income areas. In rural areas, where households are poor, traditional denominations, with their large networks of churches may hold an advantage, subsidising churches in poorer rural communities with the income from richer urban communities.

In general, little is known about how formal institutions can affect the relationship between networks and migration.

Munshi (2020) provides an effective summary of the literature on networks and migration. Networks are important in improving labor market outcomes for migrants (Munshi, 2003), though these outcomes are dependent on the size and dynamics of the network (Beaman, 2012), and can be subject to congestion effects (Tang, 2024). Blumenstock et al. (2023) separate the information and economic support provided by networks to migrants, while Dai et al. (2019) show that migration networks can have lasting effects on the quality of business networks.

Networks at origin are also affected by outmigration. Munshi and Rosenzweig (2016) and Morten (2019) study the trade-off between the potential income gains from migration, and the consumption smoothing benefits provided by the village risk-sharing network for permanent and temporary migration respectively. Meghir et al. (2022) find evidence that conversely, some forms of migration can increase the ability for origin communities to risk share. Lagakos et al. (2023) quantify the welfare effects of rural-urban migration in providing insurance for low income households.

My paper presents evidence that religious institutions can help to support migrants on entry into the city, offering a network for those with few prior connections. In rural areas, I present new evidence that religious institutions can act as a substitute for the village risk sharing

network. This suggests that the costs associated with the breakdown of the village network may be mitigated by the presence of service providing institutions.

This paper also contributes to several strands of literature within the economics of religion. Firstly, I contribute to the understanding of the economic drivers of religion. The seminal papers of Iannaccone (1992, 1998) discuss the role of stigma in stabilising religious communities in a club goods model. Glaeser and Sacerdote (2008) present a model where the social value of religion predicts correlations between education levels and religious participation. Bryan et al. (2021) provide evidence supporting the protestant work ethic - individuals in their experiment earned more income after conversion. Iyer et al. (2014a) run a survey documenting the physical services provided by religious institutions, showing that competition can increase provision. Fruehwirth et al. (2019) finds that religion reduces depression in adolescence. A recent group of papers discuss the network value of religious institutions. In his recent book Seabright (2024) describes religious institutions as platforms, bringing people together from different backgrounds for a variety of purposes. Chetty et al. (2022) find that religious institutions uniquely enable people from different social classes to interact with each other. Murphy et al. (2019) present evidence for the informational value of the network - individuals who find themselves in the same church are more likely to share agricultural information with each other. Religious groups can provide insurance, both physical (Chen, 2010; Ager and Ciccone, 2018) and psychological (Auriol et al., 2020). Chen (2010) find evidence that individuals become more religious during the Indonesian Financial crisis, and that participation in religious communities increases the ability to consumption smooth. Auriol et al. (2020) conduct a lab-in-the-field experiment in Pentecostal churches in Ghana, finding that formal insurance can substitute for religious participation, and that this substitution is owed not to the physical insurance offered by a community, but to the idea that God divinely insures faithful believers. Similar to Iyer et al. (2014b) I provide evidence at a granular level on the types of services that are provided by religious institutions. I specifically focus on network services, and other attributes of the network, complementing this data with household reports on services received and contributions made. In addition, I break down these services by denomination, and look at differential provision based on context and identity of the adherent.

Secondly, I speak to the literature on the intersection of religion and development. Much of the existing literature focuses on the mixed legacy of historical missions in colonial Africa on a range of development outcomes (income, welfare, mortality, fertility, education) (Nunn, 2010; Wantchekon et al., 2015; Valencia Caicedo, 2019; Jedwab et al., 2021; Lowes and Montero, 2021; Guirkinger et al., 2022; McCleary and Barro, 2019). A smaller number of papers examine the modern impact of religion on development. Borooah and Iyer (2005) look at the effect of caste and religion on education. Iyer (2016) study the effect of religious riots on election results. Bazzi et al. (2020) consider the response of religious schools following the introduction of public schools in Indonesia. In a similar vein, I find evidence that the provision of network goods by religious institutions act as a substitute for other providers. Religious institutions succeed in a setting with weak government services, and replace traditional village networks both in urban areas, where migrants from many ethnic groups interact, and in rural areas as village networks break down.

Finally, I provide a novel explanation for the rise of Pentecostalism in developing economies, contributing to the literature on the economic forces that underlie patterns of religious adherence. Barro and McCleary (2005, 2024) use reports from the World Value's Survey to conduct cross country comparisons on the determinants of religious conversion. In the history literature Rubin (2014) examines the effect of the spread of the printing press on the rise of Protestantism, while Becker et al. (2017) looks at the effect of the expansion of education on religious attendance in Germany. Michalopoulos et al. (2018) find evidence supporting trade as an important factor in the early spread of Islam. Corbi and Sanches (2021) find that government regulation and tax incentives can favour certain types of religious institutions. A subset of this literature establishes the effect of negative shocks on religiosity: Buser (2015) examines the effect of income on religiosity using variation in eligibility for government welfare; Sinding Bentzen (2019) find increases in conversion and conversion to Evangelical denominations in response to earthquakes; and Costa et al. (2023) find that economic downturns in Brazil led to higher local vote shares for Pentecostal candidates. The closest paper in this literature is that of Ager and Ciccone (2018), which tests the hypothesis that agricultural risk promoted the development of religious communities in the US. My paper builds migration into this story, proposing that rural-urban migration interacts

importantly with this risk sharing function to promote Pentecostalism.

The rest of the paper proceeds as follows, section 2 describes the data, section 3 describes the empirical strategy, section 4 presents results on the religious affiliation of migrants, and discusses potential mechanisms, section 5 presents results on the religious affiliation of out-migrant households, and potential mechanisms, section 6 contains a discussion of these results and concludes.

2 Data

2.1 Survey of Pentecostal Churches

Very little empirical data exists on the nature of religious institutions in developing countries (Iyer et al. (2014b) provides a notable exception). In collaboration with a team from the University of Environment and Sustainable Development in Ghana, I conduct extensive explorative surveys of a set of Pentecostal churches throughout Accra. Two communities were selected that were roughly representative of communities in Accra. In the first stage, we built a sampling frame for each community. Enumerators approached community centres, and asked administrators for full lists of the churches within the community. Enumerators walked the communities and validated these lists, adding any additional institutions they found. In the second stage, enumerators visited every Pentecostal church in the sampling frame and requested interviews with the lead pastor and/or a lead administrator within the organisation. If subjects were not available, enumerators made arrangements to follow up at another time. The final take up rate for these interviews was approximately 75%, for a total of 35 Pentecostal churches. Two enumerators sat with each subject for the interview and recorded their responses on paper surveys. Interviews were recorded to assess data quality. The survey consisted of 5 units: Institutional history, membership and attendance, services provided by the institution, church organisation and employment, and finance. Summary statistics on basic institutional information are recorded in table —. Attendance figures over time are shown in figure —.

2.2 ISSER-Northwestern-Yale Ghana Long Term Socioeconomics Panel Survey (GSPS)

Testing the relationship between migration and Pentecostalism requires longitudinal data tracking both migration movements, and religious affiliation at the denominational level. Few datasets break down denomination past the level of Protestant/Catholic level. The GSPS survey tracks denominational affiliation, including Pentecostal affiliation across its four waves (2009/2010, 2013/2014, 2017/2018, 2022/2023). The dataset also contains exten-

sive information on education, assets, income, health, consumption, networks, psychology and housing characteristics, allowing rich mapping between religious affiliation and other household economic variables. I contribute a unit in the fourth wave of the survey to elicit additional information on religious history and affiliation. The unit refines religious affiliation to the organisation level, and tracks religious history back to birth, allowing estimation to control for religion at birth, and to more closely match the timing of conversion with the timing of migration. The unit records details on the institution the household member attends, the size of the institution and the distance travelled to get there. Importantly it records information on the contributions to the religious institution, both monetary and time, and services received from the institution. This data allows comparisons on services provided by institutions between denominations, across geographies, and by individual attributes. Table — provides some basic summary statistics. The GPS survey is representative at the regional level, with 5009 households surveyed in the first wave. The panel attempts to attract migrants and split households across waves. Split households make up a significant portion of the sample, with a total of 5927 households in the fourth wave.

2.3 Supplementary datasets

The Ghana Census (2000, 2010 and 2021) releases 10% representative samples and contains information on recent migration, immigration history, and employment status. It also records religious affiliation, including a category for Pentecostal affiliation. This data provides additional information at the national level, which I use in constructing motivating facts, and in the setup of the instrument.

I use price series from the FRED database released by the US Bureau of Labor Statistics. To construct the mining exposure instrument, I use the Producer Price Index for Gold Ore. For the construction instrument I use the price of ready-mix concrete. For the manufacturing instrument I use the series for PPI private capital equipment for manufacturing industries as a measure of capital costs, and I use the yearly series for the price level of exports for Ghana, provided by the Penn World Tables.

2.4 Construction of Variables

Migration The GPS survey records migration in three different measures. It firstly records immigration information - the years living in the current community, where they were living beforehand, the reasons for migrating, and employment before and after moving. Secondly in the first three waves the survey asks for up to four migration events longer than one year in length, and migrations longer than one month in the fourth wave. The migration history unit collects information on the timing of migration, employment before and after migration, connections at destination when migrating, and reason for migrating.

Using the latter dataset, I construct a yearly migration panel. I do not use this as my primary migration measure for several reasons. Firstly, there is significant bunching of historical migration around 5 and 10 year increments, making it difficult to map migration events to pull-migration shocks, and to conversion events. Secondly, there is evidence that there is significant under-reporting of migration events. While immigration events should show up in the migration unit, this rarely occurs (in wave 3, 6184 individuals report to not being born in the community, while only 3309 report having ever lived away from the community). Additionally, the migration unit asks for the most recent migrations, preferably those since the last wave. This leads to some double reporting of migration events across waves, with conflicting reports on the timing of migration. Locations are often not reported, making it difficult to map these migration events to one another. Lastly, and critically, these migration reports combine both temporary and permanent migration - migrants may have moved with their household to their current location, migrants may have left their original household to start a new household, or family members may have lived away from their household for several years before returning.

I use the immigration unit to construct a binary measure for rural-urban migration - individual was born in a rural community, migrated, and is now living in an urban community. I use this as a secondary measure to conduct robustness checks on the migration results, and use the information it contains on employment information before and after immigration.

I use the migration panel to make use of the information it contains on reasons for migration, connections at destination when migrating, and employment information before and after

migration.

My primary measure for migration uses observed migration events from the panel. In each wave, households are asked to provide a household roster. If households report that someone is no longer belongs to their household, households are asked whether this individual is still in the community, and if not, where they migrated to. The panel then attempts to locate this individual to interview their new household. One drawback of this measure is the significant attrition of migrants. While 5695 outmigration events are recorded, only 755 of these migrants are located in the panel. This attrition will bias the results if attrition is non-random. I control for this attrition using propensity weighting of observables at baseline. This method does provide a sharp measure of out-migration. Households who report out-migration are still observed in the sample, providing a large sample size for analysis concerning these households.

Religious Affiliation The Ghana Panel records religious and denominational affiliation in each wave. In defining Pentecostal, I group together several categories which are often separated in different literatures (Pentecostal/Charismatic/Renewalist/Independent). For my purposes, the term Pentecostal is used to describe churches which are relatively *informal* - low start up costs, few formal requirements for pastors and for members, few restrictions on activities undertaken by the church; and *independent* - the organisation is independent of a large historic global denomination, the organisation has lower levels of bureaucracy, and decisions are made at a more local level. In some cases Pentecostal organisations closely resemble Protestant organisations. The Church of Pentecost, for example, claims a membership over 10% of the Ghanaian population, with origins tracing to Welsh Revival Movements in the early 1900s. They report over 18000 local assemblies across the country. While the size of this organisation surpasses any of the Protestant denominations, the organisational structure remains unique in several ways. The organisation and leadership are Ghanaian, with a much smaller presence (largely diaspora communities) internationally. Church of Pentecost branches have more independence and decision making power than branches of traditional Protestant denominations. The congregation has a greater say on the use of resources, and activities of the church. There are few training requirements for pastors in the Church of Pentecost, and local branches rely heavily on volunteers.

In the fourth wave of the survey, I collect information on the specific religious organisation that individuals belong to. I conduct robustness checks separating out these larger Pentecostal denominations.

In wave three of GSPS they introduced the category “Charismatic”. In previous waves those selecting Charismatic are likely to present in the “Pentecostal” category or the “Other Christian” category. In the first two waves I take “Other Christian” and “Pentecostal” to represent Pentecostal, though a small number of those in the “Other Christian” group would traditionally be classed as Protestant (e.g. Church of Christ). I group Anglicans, Presbyterians and Methodists as Protestant. The other groups I use are Catholic, Traditional (Ethnic) religions, and Muslim. Approximately 1.4% of the panel identify as no religion. I include them as a group in the analysis, but do not present any results related to the group. There is very little switching between Muslim affiliation and Christian affiliation. For the purpose of this paper I keep Muslim in the sample, but do not report results related to Muslim affiliation. I conduct robustness checks omitting Muslim from the panel.

Conversion I construct a set of binary variables representing conversion. $Conv_{rel}$ represents a change in reported affiliation across waves between any of the major groups discussed in the previous paragraph. $Conv_{denom}$ represents a change between any minor denominational group (for example, Methodist becoming Presbyterian). This measure is complicated by the change in definition between waves two and three. I code “Other Christian” as “Charismatic” in waves one and two for this purpose. $Conv_{Pent}$ represents a change in affiliation from any group towards Pentecostalism.

Income

3 Empirical Framework

In this section I seek to establish that migrants, who value network services, convert upon migrating, and convert towards Pentecostal denominations.

For individual i in community d at period t , the effect of migration on affiliation to Religion Rel is estimated with

$$Rel_{idt} = \beta_R Rel_{i,birth} + \beta_M Mig_{idt} + \beta_X X_{idt} + \alpha_d + \alpha_t + \varepsilon_{idt}$$

where Rel_{idt} is a dummy variable which equals 1 if individual i belongs to religion Rel at time t , $Rel_{i,birth}$ is the religion of individual i at birth, Mig_{idt} is a dummy variable which takes the value of 1 if individual i migrated between periods $t - 1$ and t , α_d is the fixed effect for the community at time 0, and α_t is the time fixed effect.

This estimation compares migrants with non-migrants from the same community at the baseline period, holding fixed individual characteristics in X_{idt} .

However, migration may be endogenous to religious choice for several reasons. There may be selection issues in that individuals who are migrants are also more likely to be Pentecostal. For example, migration will be endogenous if younger people are both more likely to demand the services provided by Pentecostal churches and to migrate to the city. Reverse causality would be a problem if conversion assists in the process of migration, if, for example, joining a Pentecostal church connected you to an inter-district network of churches. Finally, there may be unobserved variables that are correlated with both religious affiliation and migration choices. Several papers (Chen, 2010; Ager and Ciccone, 2018; Buser, 2015) have demonstrated that negative income shocks are likely to increase religiosity. Since income shocks will also affect migration choices, migration will be endogenous if there is no control for these shocks.

3.1 Instrument Design

To address these endogeneity concerns, I construct a set of pull instruments using as-good-as-random shocks to industries at potential destination districts. The instrument z_{iotk} represents

the exposure of individual i from origin district $o \in \Theta$ at time t to price shocks in commodity k in each potential destination district d . This is a market access style shift-share, where individuals who live closer to districts with high employment potential in industry k are more exposed to price shocks that affect that industry. The instrument is constructed as

$$z_{iotk} = \phi_{i(t-1)} \sum_{d \in \Theta/o} \frac{\Delta Price_{kt} \times exposure_{kd}}{\tau_{od}}$$

$\Delta Price_{kt}$ is the sum of price innovations between wave $t - 1$ and wave t , $exposure_{kd}$ is the proportion of workers in industry k that reside in district d at baseline (constructed using the 2010 Ghana Census), τ_{od} is the distance between the centroids of district o and d , and $\phi_{i(t-1)}$ is a measure of potential individual productivity.

$$\phi_{i(t-1)} = yearsedu_{i,0} \times ageweight_{i,t-1}$$

where $yearsedu_{i,0}$ is the years of education for individual i at time 0 and $ageweight_{i,t-1}$ is a measure of the likelihood of migration based on the age of the migrant at time $t - 1$, constructed using the 2010 census.

3.2 Instrument Relevance

I exploit random price innovations which affect employment opportunities in one largely rural industry - gold mining, and two urban industries - construction and manufacturing. In 2022, gold made up 47.7% of total exports in Ghana. Work in small scale and artisanal mines is extensive (the number of people working informally in small mines is estimated to be around 1.1 million, McQuilken and Hilson (2016)). Changes in the global price of gold impacts the profitability of seeking work in gold mining. This in turn induces migration, where households living nearby will be more likely to migrate following this shock. As descriptive evidence for this effect, I regress the share of individuals who report immigrating in a particular year into a gold mining district, according to the 2021 census, on lagged changes in the global gold price series from the FRED database. Results are reported in table 2, and the comovement of series can be seen in figure 2.

	Dependent variable:		
	Prop. of Migrants (1)	Number of In-Migrants (2)	(3)
Gold Price (USD)/1000	0.0001*** (0.00004)	22.045*** (1.699)	
Innovation			0.018*** (0.004)
lag(Innovation)			0.013*** (0.004)
District FE	Yes	Yes	Yes
Observations	734	734	749
Adjusted R ²	0.467	0.274	0.125
F Statistic	38.835***	17.299***	6.942***

Table 2: Migrants are Adult Males. Regression is at the year-district level.

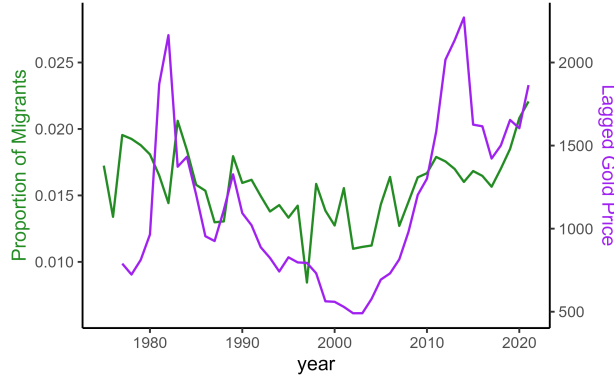


Figure 2: Caption

Secondly I use innovations in the price of construction input (FRED, 2024) multiplied by the US dollar-Ghanaian cedi exchange rate. The construction industry makes up approximately 15% of GDP in Ghana, and employs approximately 420,000 people, especially in urban areas (International Trade Administration, 2023). While approximately 7.9

Thirdly, I use innovations in the price of manufacturing inputs relative to export prices, following the analysis of Kose and Riezman (2001) in examining the effect of trade shocks in Africa. Manufacturing amounts to approximately 11

As in Madhock et al. 2024, I interact the instrument with a measure of individual productivity. The use of this second share exploits individual variation in exposure to the industry shocks. Individuals who are in their early 20s, and who have received more education are more mobile, and therefore are more exposed to the pull-migration shocks.

The two stage least squares framework is the following,

$$Rel_{ioidt} = \beta_R Rel_{i,birth} + \beta_M Mig_{ioidt} + \sum_k exp_{otk} + \beta_X X_{idt} + \alpha_o + \alpha_t + \varepsilon_{idt}$$

$$M_{iodt} = \beta_R Rel_{i,birth} + \sum_k \mu_{1,k} z_{iotk} + \sum_k exp_{otk} + \mu_x X_{iot} + \gamma_o + \gamma_t + v_{iot}$$

where $exp_{otk} = \Delta Price_{kt} \times exposure_{ko}$ is the direct exposure of district o to the commodity k shock.

First stage regressions are reported in tables —.

3.3 Instrument Exogeneity

In order for the instruments to be exogenous, it must be that the instrument only affects religious choice through migration, holding fixed controls. Recent literature in shift-shares has established that consistency can be achieved either through plausible exogeneity of the shifts (Borusyak et al., 2022) or of the shares (Goldsmith-Pinkham et al., 2020). In this section, I will argue that 1) exposure and distance shares are plausibly exogenous, i.e. that trends over time for districts with more or less exposure are the same absent of migration, and 2) that conditional on controls, shocks to employment opportunities in destination districts are randomly assigned across individuals.

I first consider the exposure of districts to shocks in commodity prices. If estimation were to occur at the district level, we would require that the equivalent of a parallel trends assumption; that trends in Pentecostalism over time across districts are the same absent of the price shock. Without controls, this is unlikely to hold. Districts closer to high density Pentecostal districts are likely to experience differential exposure over time, and districts that are more remote are likely to experience less exposure to religious pluralism. To address these issues I firstly control for region-time fixed effects, so that identification occurs by exploiting differences in the within-district exposure to industry shocks. I also control for population density, and access to Pentecostalism, by including a weighted distance measure to districts with Pentecostalism, similar to the construction of the instrument. I also control for initial percentage Pentecostal at baseline. To address concerns that exposure to price shocks is correlated across space I directly control for local district exposure to the price shock ($\Delta Price_{kt} \times exposure_{kd}$) There may also be concerns that price shocks in close by districts have spillover effects, either through trade or existing transfer networks. To address

this I control for average district income. In figure — below I show that districts with similar weights have similar growth patterns given controls.

Having established that the price-exposure measure provides plausibly random variation across districts, I transform this to an individual exposure, by multiplying by an individual productivity weight - the product of years of education at baseline and an age weight, based on the national likelihood of migrating at age a based on statistics from the 2000 Ghana census. I control directly for age and years of education level, so that identification occurs through the differential exposure of age-education individuals to the price shock. However many unobservables may be correlated with age and education that predict Pentecostal affiliation. Income, or fertility, for example, will be correlated with both age and income, and may affect religious choice. Using the approach of (Borusyak et al., 2022), the exclusion restriction then rests on the exposure shocks being as-good-as-randomly-assigned given controls and fixed effects, i.e. that exposure to district shocks does not systematically differ by individual productivity weighting ϕ_{it-1} . In figure — below I compare the distribution of shocks for individuals with high and low ageweights, and high and low education levels, after conditioning on controls and fixed effects. I directly control for income, household size, number of births and number of deaths in the household. The distributions appear very similar, so that individuals with different productivity measures are exposed to similar levels of the shock.

More formally, following Borusyak et al. (2022) I run a set of balance tests, regressing potential confounding factors, including education and age weight, against differences in the shock in a pooled cross-section, controlling for region fixed effects. If the shock is uncorrelated with these confounding factors, then we can conclude the shocks are as-good-as-randomly assigned, and that the shock is not arbitrarily picking up comovements with other individual characteristics.

4 Migration Results

Since the industry shocks I am using, especially in mining and construction will predominantly affect men, and because the majority of migration for women is for marriage or family purposes, I restrict the sample to male migrants. The estimation here is limited by restrictions in the data - though enumerators made attempts to track down migrants across waves, there is significant attrition. I adjust for this attrition using inverse probability weighting based on observables ¹. The results of the instrument regression are presented in table ??.

Columns (1) - (3) are the results for migrants who began in rural areas, and (4) - (6) are the results for migrants who began in urban areas. Male migrants from rural locations are more likely to be Pentecostal after migrating, and less likely to be Protestant. Conversely, male migrants from urban areas are less likely to be Pentecostal after migrating, with some evidence of a shift towards Protestantism.

Table 3

	<i>Dependent variable:</i>					
	<i>Pentecostal</i>	<i>Catholic</i>	<i>Protestant</i>	<i>Pentecostal</i>	<i>Catholic</i>	<i>Protestant</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Migrate</i>	0.338*** (0.127)	-0.138 (0.105)	-0.340*** (0.114)	-0.292** (0.142)	0.002 (0.075)	0.221 (0.157)
Mean	0.26	0.14	0.16	0.4	0.11	0.24
F^{Eff}	12.8003	12.8003	12.8003	7.3576	7.3576	7.3576
Region-Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,320	4,320	4,320	2,158	2,158	2,158
Adjusted R ²	0.372	0.353	0.324	0.376	0.342	0.299

A corresponding regression can be run to determine the effect of migrating on conversion - conversion between religions, conversion between denominations or conversion to Pentecostalism. The results are given in table 4. The results show no significant increase in conversion rates for migrants from either rural or urban areas.

In general the results in this section are limited by the relatively small sample size, and the attrition of migrants across waves, so an increase in conversion rate cannot be ruled out. However there may be two forces acting simultaneously which produce this null result

¹In most cases, it is known which individuals have migrated, since the household at origin reports their migration event. However, there is no information on the current religious affiliation of migrants that are not tracked down

- non-Pentecostals may migrate to the city and convert, while Pentecostals upon migrating are less likely to convert. This seems plausible, given the positive sign on the conversion to Pentecostalism regression.

Table 4

	<i>Dependent variable:</i>					
	<i>ConvAny</i>	<i>ConvDenom</i>	<i>ConvPent</i>	<i>ConvAny</i>	<i>ConvDenom</i>	<i>ConvPent</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Migrate</i>	-0.074 (0.116)	0.202 (0.163)	0.048 (0.078)	-0.093 (0.141)	-0.152 (0.143)	-0.074 (0.087)
Mean	0.2	0.27	0.06	0.15	0.24	0.02
F^{Eff}	12.7873	12.7873	12.7873	7.4413	7.4413	7.4413
Region-Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,317	4,317	4,317	2,157	2,157	2,157
Adjusted R ²	0.178	0.224	0.089	0.146	0.208	0.059

4.1 Migrant Mechanism

4.1.1 Network Services

I now seek to establish the mechanism behind this shift in religious affiliation for migrants moving from rural areas. I provide evidence to support the hypothesis that these migrants join Pentecostal churches for the network services they provide. Firstly, I demonstrate that Pentecostal churches provide economic assistance in the form of transfers significantly more than other religions and denominations.

Table 5 presents summary statistics from the religion unit in the fourth wave of the GSPS survey on both the services received from the religious institution, and contributions made. Table A.4 tests for differences in the provision of services across denominations, controlling for age, gender, household structure, income, and education levels. The significance results are reported in table 5. Pentecostals are significantly more likely than other denominations to report receiving transfers directly from their institution within the previous year. 10% of Pentecostals in urban areas report receiving at least some transfer from their institution within the last year. Pentecostals have the highest coefficient on transfers received from other members of the institution, and in businesses advertised, though these differences are not statistically significant. The second highest level of transfers is provided by religious

Denomination	N	Services Received from Institution				
		From Institution		From Network		<i>Advertise</i>
		<i>TransMoney</i>	<i>TransIn-Kind</i>	<i>TransMoney</i>	<i>TransIn-Kind</i>	
Catholic	268	0.02	0.03	0.02	0.03	0.03
Pentecostal	1420	0.06**	0.09**	0.04	0.06	0.05
Protestant	700	0.05	0.07	0.04	0.05	0.02
Muslim	550	0.01	0.02	0.00	0.01	0.04
Traditional	58	0.03	0.02	0.00	0.00	0.00

Denomination	N	Contributions to Institution				
		<i>Tithe</i>	<i>Giving</i>	<i>ReligItems</i>	<i>AttendServices</i>	<i>AttendOther</i>
Catholic	268	0.44	62.34	0.14	66.93	43.93
Pentecostal	1420	0.66***	40.66	0.14	70.99	53.09
Protestant	700	0.69***	32.12	0.23	63.40	49.54
Muslim	550	0.19	49.39	0.10	226.87***	106.89***
Traditional	58	0.02***	22.76	0.00	43.76	31.74

Table 5: Ghana Panel Wave 4 summary statistics - *TransMoney* is a binary variable which takes the value 1 if an individual received a monetary transfer within the last year, *TransIn-kind* is a binary variable which takes the value 1 if an individual received an in-kind transfer within the last year, *Advertise* is a binary variable which takes the value 1 if an individual advertised their business in their religious institution within the last year. *Tithe* is a binary variable which takes the value 1, if the individual reports giving a set portion of their income to their religious institution, *giving* is the amount of giving outside of the tithe, *ReligItems* is the amount spent on purchasing religious materials, *AttendServices* is the number of services attended during the year, and *AttendOther* is attendance at activities outside of the main service. Stars represent the significance for the difference with Catholic following regressions reported in tables A.4 and A.2, *p<0.1; **p<0.05; ***p<0.01.

institutions. There appears to be some selection into Protestant churches. After controlling for district fixed effects, the coefficient on Protestant reduces and becomes statistically insignificant. I do not include information on job and housing referrals here, since reported referrals are less than one percent in each denomination.

On the other side of the equation, Pentecostals and Protestants appear to be similarly likely to give a set proportion of their income to their religious institution (i.e., to tithe), significantly more than Catholics and non-Christians. Giving and spending outside of this tithe is relatively small, with no significant differences across denomination.

To understand the scale of these services from an institutional standpoint, table 6 reports results from the survey of 35 Pentecostal churches in Accra. Of the 32 churches that responded to the services section of the survey, 31 of them reported giving monetary transfers to their members, with a median of 12 transfers per year. Most also reported in kind transfers in terms of food and clothing. The median size of a transfer is 475 Cedi, representing approximately 1/4 of the median monthly urban household income in our sample (1870 Cedi). In kind transfers tend to be smaller, and to be given to households with lower monthly incomes. The median size of church in the sample is 110, indicating that churches report giving approximately 10% of their members a transfer within a year, in line with the GPS results. Churches report helping more members to find jobs and housing than reported by individuals

in the GSPS. Finally, churches report providing support for businesses started by their members. The reports on advertising opportunities are reflected in the GSPS data, though few people report to having attended business or skills training.

Pentecostal churches, therefore, appear to be providing network related services, especially transfers, in urban areas. Most of these transfers are centralised, and are channelled formally through the organisation. This support is supplemented by transfers that are given informally by members of the same religious institution. Notably, Pentecostal churches are significantly more likely than any other group to provide transfers through the institutional channel. These services come at a cost, with Pentecostal churches requiring greater tithing commitments.

	Number Churches	Mean	Median	Std	Min	Max
Number of Transfers						
Monetary	31	17.15	12	17.06	1	80
Food	26	17.54	10	19.08	2	80
Clothing	22	16.25	10	20.93	1	80
Size of Transfers						
Monetary	31	1749.17	475	2324.41	100	8000
Food	26	765.42	100	1484.98	50	5000
Clothing	22	577	100	1144.35	50	5000
Other Network Services						
Helped find job	28	15	7.50	26.22	2	135
Helped find housing	26	8.46	5	12.82	1	60
Business Related Services						
Business Training (Participation)	16	45.12	30	46.24	3	200
Skills Training (Participation)	16	39.31	28.50	40.66	4	170
Businesses Advertised	25	8.48	5	9.31	1	40

Table 6: Survey of 35 Pentecostal churches in Accra, Ghana: services provided in the previous year.

I now examine which migrants convert in the city, and in particular, I test if migrants with fewer connections in the city on arrival are more likely to convert. If Pentecostal churches are providing a doctrine that is particularly appealing to the type of people who migrate, we would not expect there to be differential conversion based upon connections at the destination.

To test this, I construct a year-by-year panel for migration using the self-reported migration information from the GSPS. I add a year-by-year panel for conversion, using observed conversion during the years of the Ghana Panel, and self-reports on when individuals converted away from their religion at birth. The construction of the conversion panel will miss conversions if the individual converts multiple times before 2009. The majority of individuals who convert report only converting once in their lifetime. I remove individuals from this

estimation who report converting more than once, with these conversions not appearing in the years of the panel ($< 5\%$). Self-reported migration and conversion contains significant reporting error, and in particular, significant bunching of reported migration around 5 year intervals. To address this concern, I look at the effect of migration on conversion in the 5 years following the migration event. I run a fixed effects regression of the following form:

$$Conv_{idt} = \beta_R Rel_{i,birth} + \beta_{M_1} Mig_{NoConn,idt} + \beta_{M_2} Mig_{Conn,idt} + \beta_X X_{idt} + \alpha_i + \alpha_t + \varepsilon_{idt}$$

The results are reported in table . The results demonstrate that in periods where an individual migrates with no connections, they are significantly more likely to convert. Conversely, there is no significant effect of migrating with connections on conversion.

	<i>Dependent variable:</i>			
	<i>Convert</i> (1)	<i>Convert_{Pent}</i> (2)	<i>Convert</i> (3)	<i>Convert_{Pent}</i> (4)
Know Relative/Acquaintance	-0.002 (0.010)	0.009 (0.010)	-0.008 (0.010)	0.004 (0.009)
Know No One	0.035* (0.018)	0.020 (0.014)	0.037** (0.016)	0.023* (0.013)
Mean	0.19	0.26	0.19	0.26
Ind FE	No	No	Yes	Yes
Observations	128,835	128,835	128,835	128,835
Adjusted R ²	0.069	0.063	0.217	0.227

Table 7: Ghana Panel year by year regression using self-reported migration events. *Convert* is conversion in the 5 year period after migration. *Convert_{Pent}* is conversion to Pentecostalism. Errors clustered at the EA level.*p<0.1; **p<0.05; ***p<0.01

Consistent with these results are several remaining hypotheses. It may be the case that upon migrating, the migrant increases their income, and this increase in income is the factor that leads to conversion. This is inconsistent, however, with the evidence presented for migrants with fewer connections, who are less likely to be employed upon migrating. Table — shows that men who are unemployed upon migrating are more likely to convert. I also conduct a robustness test in table — comparing the effect of income and instrument price changes on conversion for migrants vs non-migrants. I find no evidence for a differential conversion effect of income between migrants and non-migrants. It may be the case that migrants, and especially disenfranchised migrants who have not found employment in the city are especially attracted to the doctrines presented in Pentecostal churches, in providing

hope and meaning to their lives. This story would be consistent with the evidence offered by Binzel and Carvalho (2017), where unemployed males in Egypt disappointed with their job market outcomes shift their time use towards religious practices. To test this, I look at the questions related to religious intensity. If disenfranchised migrants are simply becoming more religious, we should see an increase in religiosity outside of interactions with the network, when compared to employed converts. Given religious intensity is only observed at the fourth wave, results here are limited. [could also look at transfers, or interactions with community, or hopelessness/depressed measures] It may also be the case that migrants do value network services, but it is the social, rather than economic component of the network that they value. In this case, there should be no difference in conversion based on economic outcomes in the city. Holding fixed network connections, employment outcomes should not predict conversion.

5 Out-Migration

In this section, I investigate the effect of out-migration on the religious affiliation of households at origin. Outmigration may affect religious affiliation through several channels - a household with an out-migrant may be exposed to a new religion product for the first time, a household with an out-migrant may join the religion of the migrant as a form of support, a household may increase their religiosity as a form of divine investment in the migrant. I test these against the hypothesis that outmigration breaks down the village network as in Morten (2019), and the breaking down of the village network increases the demand for alternative sources of network goods, which are provided by religious institutions.

I run a household equivalent regression to the one used in the section above.

$$Rel_{hodt} = \beta_R Rel_{h,birth} + \beta_M Mig_{hodt} + \sum_k exp_{otk} + \beta_X X_{idt} + \alpha_o + \alpha_t + \varepsilon_{idt}$$

$$M_{iodt} = \beta_R Rel_{i,birth} + \sum_k \mu_{1,k} z_{iotk} + \sum_k exp_{otk} + \mu_x X_{iot} + \gamma_o + \gamma_t + v_{iot}$$

Rel_{hodt} is the number of household members in household h that belong to religion Rel at time t , $Rel_{h,birth}$ is the modal religion of the adult household members at birth, Mig_{hodt} is the number of out-migrants from household h that migrated between periods $t - 1$ and t , α_d is the fixed effect for the community at time 0, and α_t is the time fixed effect.

The instrument is defined using the same industry shocks as in the migration section. The instrument is modified to represent exposure at the household level. The same arguments for exogeneity stand as discussed. The instrument expression becomes

$$z_{hotk} = \phi_{h,0} \sum_{d \in \Theta/o} \frac{\Delta Price_{kt} \times exposure_{kd}}{\tau_{od}}$$

$$\phi_{h,0} = yearsedu_{h,0} \times ageweight_{h,0}$$

where $yearsedu_{h,0}$ is the total years of education for household h at baseline and $ageweight_{h,0}$ is the average age of adults in household h at baseline.

The results of this regression are given in table 8.

Table 8

	<i>Dependent variable:</i>			
	<i>Pentecostal</i>	<i>Catholic</i>	<i>Protestant</i>	<i>Ethnic</i>
	(1)	(2)	(3)	(4)
Num_{Mig}	0.060** (0.030)	0.106*** (0.032)	0.048** (0.024)	0.149** (0.062)
‘malemig(fit)’	−0.381** (0.178)	−0.060 (0.085)	0.091 (0.182)	0.102 (0.078)
Mean	1.166	0.563	0.554	0.287
F^{Eff}	37.9122	37.9122	37.9122	37.9122
HH Control	Yes	Yes	Yes	Yes
Observations	6,860	6,860	6,860	6,860
Adjusted R^2	0.660	0.703	0.632	0.574

These results show that households with outmigrants are less likely to have Catholic members as a result. Though not significant, Pentecostal also appears to decline, in favor of Protestantism. Running the regression on conversion obtains the results given in table 9. Out migration increases conversion rates between religions and denominations. While conversion rates to Pentecostalism increase, this is evidently less than the conversion away which leads to a net positive for Protestant affiliation.

Table 9

	<i>Dependent variable:</i>		
	<i>Conv_{Rel}</i>	<i>Conv_{Denom}</i>	<i>Conv_{Pent}</i>
	(1)	(2)	(3)
Mig_{Out}	0.280*** (0.065)	0.307*** (0.076)	0.154*** (0.047)
Mean	0.458	0.603	0.141
F^{Eff}	37.9122	37.9122	37.9122
Observations	6,860	6,860	6,860
Adjusted R^2	0.177	0.208	0.016

5.0.1 Out-Migrant Mechanism

In this section, I provide evidence that households are converting in response to a weakening village network, and that Protestant churches are more effective service providers in rural areas.

Table 10 compares services provided by religious institutions in rural areas in the Ghana

Panel, with stars representing differences in services provided, or contributions made compared to Catholic affiliated households. Tables A.3 and ?? report the regressions.

Denomination	N	Services Received from Institution				
		From Institution		From Network		<i>Advertise</i>
		<i>TransMoney</i>	<i>TransIn-Kind</i>	<i>TransMoney</i>	<i>TransIn-Kind</i>	
Catholic	843	0.03	0.03	0.01	0.02	0.02
Pentecostal	2102	0.05	0.04	0.03***	0.03	0.03
Protestant	1053	0.04	0.03	0.02	0.02	0.03
Muslim	1189	0.01	0.03	0.00	0.02*	0.02
Traditional	475	0.01	0.01	0.01	0.01	0.01

Denomination	N	Contributions to Institution				
		<i>Tithe</i>	<i>Giving</i>	<i>ReligItems</i>	<i>AttendServices</i>	<i>AttendOther</i>
Catholic	843	0.46	13.23	0.13	69.05	46.49
Pentecostal	2102	0.61***	14.96	0.10	66.42	48.53***
Protestant	1053	0.64***	21.01	0.12	63.86	45.07
Muslim	1189	0.17***	11.44	0.09	242.48***	114.09***
Traditional	475	0.03***	0.65	0.00**	43.47***	34.60 ***

Table 10: Ghana Panel Wave 4 summary statistics - *TransMoney* is a binary variable which takes the value 1 if an individual received a monetary transfer within the last year, *TransIn-kind* is a binary variable which takes the value 1 if an individual received an in-kind transfer within the last year, *Advertise* is a binary variable which takes the value 1 if an individual advertised their business in their religious institution within the last year. *Tithe* is a binary variable which takes the value 1, if the individual reports giving a set portion of their income to their religious institution, *giving* is the amount of giving outside of the tithe, *ReligItems* is the amount spent on purchasing religious materials, *AttendServices* is the number of services attended during the year, and *AttendOther* is attendance at activities outside of the main service. Stars represent the significance for the difference with Catholic following regressions reported in tables A.4 and A.2, *p<0.1; **p<0.05; ***p<0.01.

The results demonstrate no significant differences in the transfers received between any religious group, except that Pentecostals appear to be more likely to receive monetary transfers from their network compared to Catholics. The number of transfers provided to rural households is generally smaller. It seems, therefore, that the Christian denominations resemble each other more strongly in rural areas in terms of the services being provided. Conversely, differences in giving appear even more stark, with Pentecostals and Protestants both much more likely to tithe than their Catholic counterparts. Pentecostal and Protestant groups appear virtually indistinguishable in rural areas.

I now consider what the effect of out-migration is for the relationship between the out-migrant household and their village. The Ghana Panel gathers information both on general trust in the village, and asks specifically about other survey households and their interactions with them. Table ?? reports the effect of migration on the household's trust in the village network. Out migration lowers trust the household trust in the village network, consistent with the story of the breakdown of village networks with migration.

Similarly table ?? reports the change in network interactions following outmigration. Out-migration lowers the number of other village households that the out-migrant household is

willing to help, or to share information with.

In both of these cases, important network services, including information are lost as out-migration occurs. These type of services are provided by religious institutions, suggesting increased religiosity or changed religion may be beneficial for these out-migrant households. Table — reports the results of a regression comparing religiosity for out-migrant and non out-migrant households.

6 Discussion

7 Conclusion

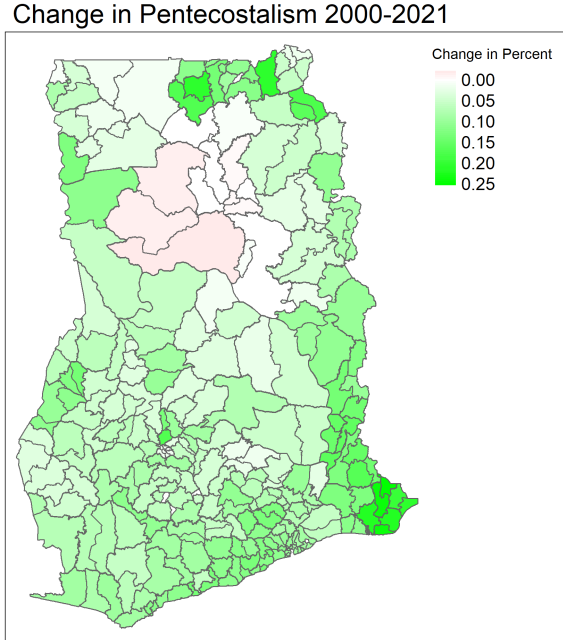


Figure A.1: Ghana Census 2000,2021

A A nice appendix

Table A.1

	<i>Dependent variable:</i>				
	<i>TransMoney</i>	<i>TransIn-kind</i>	<i>TransMoney</i>	<i>TransIn-kind</i>	Advertise
	(1)	(2)	(3)	(4)	(5)
Muslim	−0.001 (0.022)	−0.006 (0.020)	−0.009 (0.018)	−0.014 (0.017)	−0.027 (0.020)
Protestant	0.013 (0.018)	0.008 (0.016)	0.001 (0.015)	0.007 (0.014)	−0.0004 (0.017)
Pentecostal	0.036** (0.017)	0.032** (0.015)	0.014 (0.014)	0.016 (0.013)	0.024 (0.016)
Traditional	0.027 (0.042)	−0.013 (0.034)	−0.019 (0.035)	−0.013 (0.029)	−0.023 (0.039)
Constant	0.010 (0.121)	−0.017 (0.096)	0.003 (0.100)	−0.044 (0.082)	−0.041 (0.113)
Mean	0.04	0.05	0.03	0.03	0.04
District FE	Yes	Yes	Yes	Yes	Yes
Observations	2,408	3,056	2,407	3,056	2,410
R ²	0.094	0.066	0.122	0.070	0.088
F Statistic	1.461***	1.207**	1.955***	1.295***	1.358***

Table A.2

	<i>Dependent variable:</i>				
	<i>Tithe</i>	<i>Giving</i>	<i>ReligItems</i>	<i>Services</i>	<i>Activity</i>
	(1)	(2)	(3)	(4)	(5)
Muslim	−0.090* (0.053)	45.585 (37.884)	−0.038 (0.088)	125.492*** (7.954)	85.871*** (7.469)
Protestant	0.227*** (0.042)	−40.414 (30.534)	0.089 (0.071)	2.508 (6.411)	3.549 (6.020)
Pentecostal	0.230*** (0.040)	−22.300 (28.428)	0.015 (0.066)	9.471 (5.969)	6.948 (5.605)
Traditional	−0.360*** (0.088)	−47.859 (65.233)	−0.148 (0.151)	−9.099 (13.697)	2.795 (12.861)
Constant	0.305 (0.265)	−7.763 (183.460)	−0.291 (0.424)	3.336 (38.521)	−18.797 (36.171)
Mean	0.54	67.63	3.15	95.3	NaN
District FE	Yes	Yes	Yes	Yes	Yes
Observations	1,861	3,056	3,056	3,056	3,056
R ²	0.301	0.068	0.070	0.436	0.206
F Statistic	4.571***	1.259**	1.297***	13.278***	4.469***

Table A.3

	<i>Dependent variable:</i>				
	<i>TransMoney</i>	<i>TransIn-kind</i>	<i>TransMoney</i>	<i>TransIn-kind</i>	Advertise
	(1)	(2)	(3)	(4)	(5)
Muslim	−0.007 (0.009)	0.008 (0.010)	0.002 (0.006)	0.019** (0.009)	−0.003 (0.008)
None	−0.020 (0.013)	−0.018 (0.014)	−0.003 (0.009)	−0.019 (0.013)	−0.016 (0.011)
Other	−0.016 (0.057)	−0.011 (0.061)	0.008 (0.040)	0.005 (0.054)	−0.009 (0.049)
Protestant	−0.005 (0.008)	−0.004 (0.008)	−0.004 (0.006)	−0.006 (0.007)	0.003 (0.007)
Renewalist	0.009 (0.007)	0.008 (0.007)	0.013*** (0.005)	0.010 (0.007)	0.006 (0.006)
Traditional	−0.007 (0.011)	−0.002 (0.011)	0.012 (0.007)	0.005 (0.010)	0.004 (0.009)
Constant	−0.002 (0.080)	−0.030 (0.085)	−0.003 (0.056)	−0.029 (0.075)	0.0002 (0.068)
Mean	0.03	0.03	0.01	0.02	0.02
District FE	Yes	Yes	Yes	Yes	Yes
Observations	5,853	5,853	5,853	5,853	5,853
R ²	0.060	0.074	0.056	0.081	0.057
F Statistic	1.547***	1.949***	1.439***	2.157***	1.467***

Table A.4

	<i>Dependent variable:</i>				
	<i>Tithe</i>	<i>Giving</i>	<i>ReligItems</i>	<i>Services</i>	<i>Activity</i>
	(1)	(2)	(3)	(4)	(5)
Muslim	−0.166*** (0.032)	2.135 (4.403)	−0.046 (0.042)	134.280*** (4.931)	67.124*** (4.730)
None	−0.363*** (0.043)	−1.953 (6.419)	−0.120** (0.061)	−27.268*** (7.188)	−19.894*** (6.895)
Other	−0.148 (0.251)	−6.552 (27.440)	−0.050 (0.260)	49.638 (30.729)	25.645 (29.475)
Protestant	0.078*** (0.027)	4.059 (3.800)	−0.032 (0.036)	1.240 (4.256)	5.576 (4.082)
Renewalist	0.135*** (0.024)	4.072 (3.317)	−0.019 (0.031)	6.068 (3.715)	9.611*** (3.563)
Traditional	−0.254*** (0.036)	−7.527 (5.050)	−0.101** (0.048)	−30.201*** (5.655)	−18.882*** (5.424)
Constant	0.022 (0.212)	−17.923 (38.099)	2.946*** (0.361)	39.608 (42.666)	46.482 (40.925)
Mean	0.03	0.03	0.01	0.02	0.02
District FE	Yes	Yes	Yes	Yes	Yes
Observations	3,379	5,853	5,853	5,853	5,853
R ²	0.353	0.082	0.066	0.497	0.227
F Statistic	7.620***	2.180***	1.715***	24.104***	7.160***

References

- Ager, P. and Ciccone, A. (2018). Agricultural risk and the spread of religious communities. *Journal of the European Economic Association*, 16(4):1021–1068.
- Auriol, E., Lassebie, J., Panin, A., Raiber, E., and Seabright, P. (2020). God insures those who pay? formal insurance and religious offerings in ghana. *The Quarterly Journal of Economics*, 135(4):1799–1848.
- Barro, R. J. and McCleary, R. M. (2005). Which countries have state religions? *The Quarterly Journal of Economics*, 120(4):1331–1370.
- Barro, R. J. and McCleary, R. M. (2024). Religious conversion across countries. Technical report, National Bureau of Economic Research.
- Bazzi, S., Hilmy, M., and Marx, B. (2020). Religion, education, and the state. Technical report, National Bureau of Economic Research.
- Beaman, L. A. (2012). Social networks and the dynamics of labour market outcomes: Evidence from refugees resettled in the us. *The Review of Economic Studies*, 79(1):128–161.
- Becker, S. O., Nagler, M., and Woessmann, L. (2017). Education and religious participation: city-level evidence from germany’s secularization period 1890–1930. *Journal of Economic Growth*, 22(3):273–311.
- Binzel, C. and Carvalho, J.-P. (2017). Education, social mobility and religious movements: The islamic revival in egypt. *The Economic Journal*, 127(607):2553–2580.
- Blumenstock, J. E., Chi, G., and Tan, X. (2023). Migration and the value of social networks. *Review of Economic Studies*, page rdad113.
- Borooah, V. K. and Iyer, S. (2005). Vidya, veda, and varna: The influence of religion and caste on education in rural india. *The Journal of Development Studies*, 41(8):1369–1404.
- Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of economic studies*, 89(1):181–213.

- Bryan, G., Choi, J. J., and Karlan, D. (2021). Randomizing religion: the impact of protestant evangelism on economic outcomes. *The Quarterly Journal of Economics*, 136(1):293–380.
- Buser, T. (2015). The effect of income on religiousness. *American Economic Journal: Applied Economics*, 7(3):178–95.
- Chen, D. L. (2010). Club goods and group identity: Evidence from islamic resurgence during the indonesian financial crisis. *Journal of political Economy*, 118(2):300–354.
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebe, J., Hendren, N., Fluegge, R. B., Gong, S., Gonzalez, F., Grondin, A., Jacob, M., et al. (2022). Social capital ii: determinants of economic connectedness. *Nature*, 608(7921):122–134.
- Corbi, R. and Sanches, F. A. M. (2021). The political economy of pentecostalism: A dynamic structural analysis. Technical report.
- Costa, F., Marcantonio, A., and Rocha, R. (2023). Stop suffering! economic downturns and pentecostal upsurge. *Journal of the European Economic Association*, 21(1):215–250.
- Dai, R., Mookherjee, D., Munshi, K., Zhang, X., et al. (2019). The community origins of private enterprise in china. Technical report, Boston University-Department of Economics.
- Fruehwirth, J. C., Iyer, S., and Zhang, A. (2019). Religion and depression in adolescence. *Journal of Political Economy*, 127(3):1178–1209.
- Glaeser, E. L. and Sacerdote, B. I. (2008). Education and religion. *Journal of Human Capital*, 2(2):188–215.
- Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.
- Guirking, C., Villar, P., et al. (2022). Pro-birth policies, missions and fertility: historical evidence from congo. *Development Finance and Public Policies*.
- Iannaccone, L. R. (1992). Sacrifice and stigma: Reducing free-riding in cults, communes, and other collectives. *Journal of political economy*, 100(2):271–291.

- Iannaccone, L. R. (1998). Introduction to the economics of religion. *Journal of economic literature*, 36(3):1465–1495.
- Iyer, S. (2016). The new economics of religion. *Journal of Economic Literature*, 54(2):395–441.
- Iyer, S., Velu, C., and Mumit, A. (2014a). Communication and marketing of services by religious organizations in india. *Journal of Business Research*, 67(2):59–67.
- Iyer, S., Velu, C., and Weeks, M. (2014b). Divine competition: Religious organisations and service provision in india.
- Jedwab, R., zu Selhausen, F. M., and Moradi, A. (2021). Christianization without economic development: Evidence from missions in ghana. *Journal of Economic Behavior & Organization*, 190:573–596.
- Kose, M. A. and Riezman, R. (2001). Trade shocks and macroeconomic fluctuations in africa. *Journal of development Economics*, 65(1):55–80.
- Lagakos, D., Mobarak, A. M., and Waugh, M. E. (2023). The welfare effects of encouraging rural–urban migration. *Econometrica*, 91(3):803–837.
- Lowes, S. and Montero, E. (2021). The legacy of colonial medicine in central africa. *American Economic Review*, 111(4):1284–1314.
- McCleary, R. M. and Barro, R. J. (2019). *Protestants and catholics and educational investment in Guatemala*. Springer.
- McQuilken, J. and Hilson, G. (2016). Artisanal and small-scale gold mining in ghana. *Evidence to inform an ‘action dialogue’*. IIED, London.
- Meghir, C., Mobarak, A. M., Mommaerts, C., and Morten, M. (2022). Migration and informal insurance: Evidence from a randomized controlled trial and a structural model. *The Review of Economic Studies*, 89(1):452–480.
- Michalopoulos, S., Naghavi, A., and Prarolo, G. (2018). Trade and geography in the spread of islam. *The Economic Journal*, 128(616):3210–3241.

- Morten, M. (2019). Temporary migration and endogenous risk sharing in village india. *Journal of Political Economy*, 127(1):1–46.
- Munshi, K. (2003). Networks in the modern economy: Mexican migrants in the us labor market. *The Quarterly Journal of Economics*, 118(2):549–599.
- Munshi, K. (2020). Social networks and migration. *Annual Review of Economics*, 12(1):503–524.
- Munshi, K. and Rosenzweig, M. (2016). Networks and misallocation: Insurance, migration, and the rural-urban wage gap. *American Economic Review*, 106(01):46–98.
- Murphy, D. M., Nourani, V., and Lee, D. R. (2019). Chatting at church: Information diffusion through religious networks. *Review of Economics and Statistics*, pages 1–46.
- Nunn, N. (2010). Religious conversion in colonial africa. *American Economic Review*, 100(2):147–152.
- Rubin, J. (2014). Printing and protestants: an empirical test of the role of printing in the reformation. *Review of Economics and Statistics*, 96(2):270–286.
- Seabright, P. (2024). *The Divine Economy: How Religions Compete for Wealth, Power, and People*. Princeton University Press.
- Sinding Bentzen, J. (2019). Acts of god? religiosity and natural disasters across subnational world districts. *The Economic Journal*, 129(622):2295–2321.
- Tang, Y. (2024). To follow the crowd? benefits and costs of migrant networks.
- Valencia Caicedo, F. (2019). The mission: Human capital transmission, economic persistence, and culture in south america. *The Quarterly Journal of Economics*, 134(1):507–556.
- Wantchekon, L., Klašnja, M., and Novta, N. (2015). Education and human capital externalities: evidence from colonial benin. *The Quarterly Journal of Economics*, 130(2):703–757.